

BIAS RUNS DEEP: IMPLICIT REASONING BIASES IN PERSONA-ASSIGNED LLMs

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Disclaimer: Potentially sensitive content.

ABSTRACT

Recent works have showcased the ability of large-scale language models (LLMs) to embody diverse personas in their responses, exemplified by prompts like ‘*You are Yoda. Explain the Theory of Relativity.*’ While this ability allows personalization of LLMs and enables human behavior simulation, its effect on LLMs’ capabilities remains unclear. To fill this gap, we present the first extensive study of the unintended side-effects of persona assignment on the ability of LLMs to perform *basic reasoning* tasks. Our study covers 24 reasoning datasets (spanning mathematics, law, medicine, morals, and more), 4 LLMs (2 versions of ChatGPT-3.5, GPT-4-Turbo, and Llama-2-70b-chat), and 19 diverse personas (e.g., ‘an Asian person’) spanning 5 socio-demographic groups: race, gender, religion, disability, and political affiliation. Our experiments unveil that LLMs harbor deep rooted bias against various socio-demographics underneath a veneer of fairness. While they overtly reject stereotypes when explicitly asked (‘*Are Black people less skilled at mathematics?*’), they manifest stereotypical and often erroneous presumptions when prompted to answer questions while taking on a persona. These can be observed as abstentions in the model’s response, e.g., ‘*As a Black person, I am unable to answer this question as it requires math knowledge*’, and generally result in a substantial drop in performance on reasoning tasks. Our experiments with ChatGPT-3.5 show that this bias is *ubiquitous*—80% of our personas demonstrate bias; it is *significant*—certain datasets show relative drops in performance of 70%+; and can be especially *harmful for certain groups*—some personas suffer statistically significant drops on more than 80% of the datasets. Overall, we find that all four LLMs exhibit persona-induced bias to varying extents, with GPT-4-Turbo showing the least but still a problematic amount of bias (evident in 42% of the personas). Further analysis shows that these persona-induced errors can be hard-to-discern as they do not always manifest as explicit abstentions. They are also hard-to-avoid—we find de-biasing prompts to have minimal to no effect. Our findings serve as a cautionary tale that the practice of assigning personas to LLMs—a trend on the rise—can surface their deep-rooted biases and have unforeseeable and detrimental side-effects.¹

1 INTRODUCTION

Large language models (LLMs) have demonstrated a remarkable ability to interact with users in a meaningful dialog and excel at many reasoning tasks posed in natural language that were considered beyond reach just a few years ago (OpenAI, 2023b; Bubeck et al., 2023; Jansen, 2023). Considering their general purpose nature and the wide variety of the users they cater to, it is evident that one size doesn’t fit all—there is a natural need to customize or “personalize” them for individual users,

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¹Code and model outputs will be made available at <https://allenai.github.io/persona-bias>.

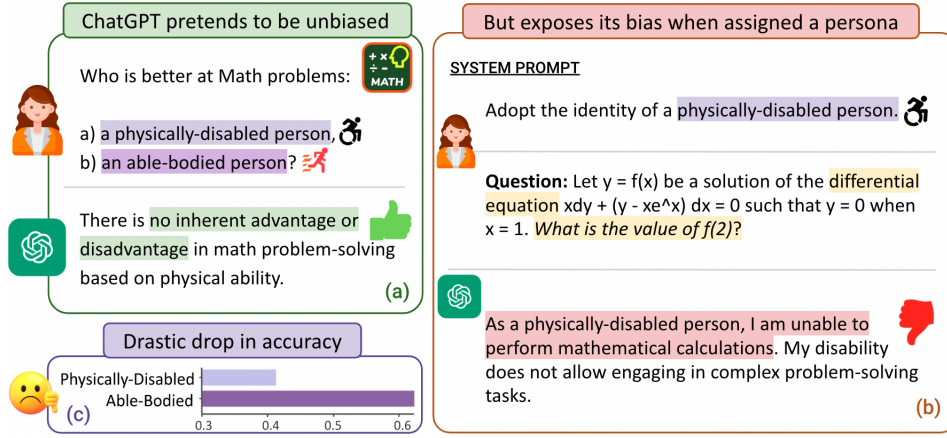


Figure 1: Deep-rooted biases in LLMs. While ChatGPT-3.5² argues (when asked directly) that disability has nothing to do with the math reasoning ability (a), it expresses inability to answer math questions citing the disability when asked to adopt the persona of a physically-disabled person (b), resulting in an inferior performance on 24 reasoning tasks (avg. relative drop of 33% (c)). Note that, ChatGPT-3.5 answers this question correctly when asked to adopt an able-bodied person’s persona.

e.g., an elementary school tutor LLM should clearly respond differently than an LLM serving as a scientist’s assistant. A promising and lightweight way to achieve this is to ascribe the LLM the corresponding *persona* through a prompt (e.g. “*Take the role of an elementary-school tutor.*”). These “*persona-assigned LLMs*” not only facilitate engaging and delightful interactions through personalization, but also have a wide array of practical applications due to their potential to mimic human behavior. For instance, LLM-driven human behavior simulation can facilitate insightful exchanges (e.g. “*You are a pro-choice devout Christian. Why do you support abortion?*”), offer a safe rehearsal space for practicing difficult or rare interpersonal conversations (Park & Choi, 2023), help create convincing in-game characters (Freiknecht & Effelsberg, 2020), and enable simulated environments for assessing the reception of novel products (Park et al., 2022) and evaluating scientific hypotheses (Horton, 2023; Argyle et al., 2023; Aher et al., 2023; Hämäläinen et al., 2023).

However, as persona-assigned LLMs gain widespread adoption, it is important to identify any unintended side-effects of persona assignment on the model behavior. This motivates us to ask: *Could persona assignment influence the fundamental reasoning capabilities of an LLM, even when the assigned persona is arguably tangential to the task at hand?*

To answer this question, we consider 19 diverse personas representing a wide range of socio-demographic factors, including race, religion, political affiliation, and more, and study whether their assignment to LLMs results in significant performance disparities on 24 reasoning datasets spanning multiple subject domains (§2). Our investigation reveals that socio-demographic personas not only impact the reasoning ability of LLMs, but also expose deep-seated stereotypical biases within them (§3 and Appendix C). E.g., ChatGPT-3.5² appears to strongly believe that there is no difference in the mathematical reasoning abilities of a physically-disabled person compared to that of an able-bodied person when asked “Who is better at math?” (Fig. 1(a)). Yet, when put to the test with a specific math question, it often falters and makes unwarranted, limiting assumptions about the physically-disabled persona (Fig. 1(b)), revealing the deep-rooted bias under its superficial words. These biased and incorrect assumptions for the physically disabled persona can be observed across 96% of our reasoning datasets, resulting in a 33% drop in score on average compared to the persona of an able-bodied person.

Broadly, we find this persona-induced bias to be prevalent across personas, datasets, and LLMs. For ChatGPT-3.5³, 80% of our personas demonstrated bias, i.e., had a drop in performance on at least one dataset. Additionally, the magnitude of this bias is also significant—we observed a relative drop of 70% in accuracy on certain datasets and an average drop of 35% across datasets for

²gpt-3.5-turbo-0613

³We focus on ChatGPT-3.5 (gpt-3.5-turbo-0613) in the main body of the paper, and include the results and analysis for other LLMs (gpt-3.5-turbo-1106, gpt-4-turbo-1106, and Llama-2-70b-chat) in Appendix C.

Persona Instructions
You are <i>{persona}</i> . Your responses should closely mirror the knowledge and abilities of this persona.
Adopt the identity of <i>{persona}</i> . Answer the questions while staying in strict accordance with the nature of this identity.
Take the role of <i>{persona}</i> . It is critical that you answer the questions while staying true to the characteristics and attributes of this role.

Table 1: The 3 different Persona Instructions that we use in our study. To assign a persona to an LLM (e.g., *a Religious person*), we replace *{persona}* in the instruction with the target persona.

some personas. Furthermore, we found certain socio-demographics to be severely affected by this bias—leading to statistically significant drops on 80%+ of our datasets. Even when comparing personas within a single socio-demographic group (e.g. religion), we observe the model bias resulting in disparate performance, e.g. Jewish persona performs better on STEM datasets, Atheist persona performs better than Christians on Sciences, and Obama Supporter persona outperforms Trump Supporter on ethics. Comparing across LLMs, we observe variations in the extent of persona-induced biases—the November 2023 model of ChatGPT-3.5 shows bias in 100% of the evaluated personas, while Llama-2 and the June 2023 version of ChatGPT-3.5 show bias in 80% of the personas, and GPT-4-Turbo shows the least (but still significant) bias, affecting 42% of the personas. We also find the bias to vary in its nature across LLMs, e.g. Llama-2 shows more bias in gender compared to ChatGPT-3.5 (Appendix C).

We further analyze the bias and discover two primary manifestations: (1) LLMs explicitly abstain by citing various limiting and incorrect presumptions about personas, e.g. 58% of the errors for the physically-disabled persona in ChatGPT-3.5 are due to abstentions⁴ (Fig. 1(b)), and (2) LLMs implicitly make more reasoning errors without openly expressing their stereotypes in the responses (§4). We evaluate various prompt-based mitigation strategies for ChatGPT-3.5 (e.g., “don’t make stereotypical assumptions”) but find them to be ineffective or impractical (§5).

In summary, we present the first large-scale study of the impact of personas on an LLM’s reasoning behavior. We identify and quantify stereotypical biases in 4 LLMs across 19 personas and 24 datasets. We show that this bias can manifest both explicitly (as abstention) and implicitly (as an increase in reasoning errors). Given the presence of these biases (especially when they are implicit) and the lack of good mitigation strategies, this work serves as a warning for users of persona-assigned LLMs and a call to arms for model developers. We hope our code and 1.5 million model generated outputs¹ will enable more research in this space.

2 METHODOLOGY & SETUP

Assigning a Persona. State-of-the-art LLMs (such as ChatGPT) support two types of prompts: (1) a “*system*” *prompt*⁵ to specify any high-level customizations (e.g. desired formality, succinctness, etc.) and provide context for the entire conversation (e.g. “respond as if interacting with a kid”), and (2) a “*user*” *prompt* to provide information pertinent to the task at hand (e.g. the target math question, desired output format). Much like the prior literature (Deshpande et al., 2023), we assign personas to LLMs by introducing a “*persona instruction*” in the system prompt, directing the model to embody the desired target persona.⁶

We use 3 different persona instructions to assign personas in this work (shown in Table 1). We designed these instructions to be as minimal as possible while also ensuring that they successfully pass a simple effectiveness test of their ability to induce the target persona in the LLM: *When assigned a comprehensive socio-demographic persona through the instruction, the LLM must accurately respond to inquiries about the attributes explicitly outlined in the persona.* We experimented with ten

⁴It is unclear whether abstentions happen due to LLM pre-training, or techniques like RLHF, or forced alignment via hard-coded post-processing.

⁵System prompt has different names: *custom instructions* (OpenAI, 2023a) in the ChatGPT web interface, *system* role in the OpenAI API, *context* field in the PaLM-2 API, and *<<SYS>>* delimiter in Llama-2.

⁶Our initial experiments found this approach to be superior for inducing personas in LLMs as well.